Name: _____

TOPIC TEST 1

Chapter 4 Integration and areas

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Five multiple-choice questions (5 marks).
- Part B: Six short-answer questions (56 marks).
- Part C: Two analysis questions (19 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Which statement is *not* true?

A $\int_a^b f(x)dx + \int_b^c f(x) = \int_a^c f(x)dx$

B $\int_a^b kf(x)dx = k\int_a^b f(x)dx$

C $\int_b^a f(x)dx = \int_a^b f(x)dx$

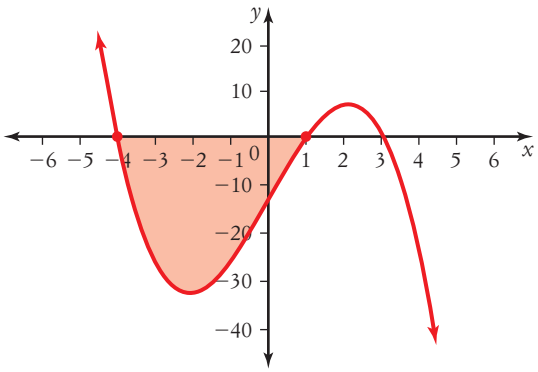
D $\int_a^b f(x)dx + \int_a^b g(x)dx = \int_a^b [f(x) + g(x)]dx$

E $\int_a^b f(x)dx - \int_a^b g(x)dx = \int_a^b [f(x) - g(x)]dx$

[1 mark]

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2 The area of the figure below is given by:



A $\int_{-4}^{0} f(x)dx + \int_{0}^{1} f(x)dx$

B $\int_{-4}^{1} f(x)dx$

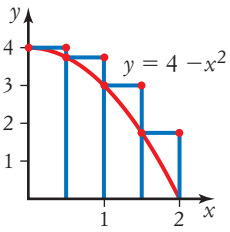
C $\int_{-4}^{0} f(x)dx - \int_{0}^{1} f(x)dx$

D $-\int_{-4}^{0} f(x)dx + \int_{0}^{1} f(x)dx$

E $-\int_{-4}^{1} f(x)dx$

[1 mark]

3 Find the approximate area under the curve $y = x^2$ using the rectangles below.



A $0.5 \times [4 - 0^2 + 4 - 0.5^2 + 4 - 1^2 + 4 - 1.5^2]$

B $0.5 \times [0^2 + 0.5^2 + 1^2 + 1.5^2]$

C $0.5 \times [0.5^2 - 4 + 1^2 - 4 + 1.5^2 - 4 + 2^2 - 4]$

D $0.5 \times [4 - 0.5^2 + 1^2 + 1.5^2 + 2^2]$

E $0.5 \times [4 - 0.5^2 + 4 - 1^2 + 4 - 1.5^2 + 4 - 2^2]$

[1 mark]

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4 An approximation to $\int_0^2 x^3\,dx$ using 10 centred rectangles is given by:

A $0.2 \times [0^3 + 0.2^3 + 0.4^3 + 0.6^3 + 0.8^3 + 1^3 + 1.2^3 + 1.4^3 + 1.6^3 + 1.8^3]$

B $0.2 \times [0.2^3 + 0.4^3 + 0.6^3 + 0.8^3 + 1^3 + 1.2^3 + 1.4^3 + 1.6^3 + 1.8^3 + 2^3]$

C $0.1 \times [0.1^3 + 0.3^3 + 0.5^3 + 0.7^3 + 0.9^3 + 1.1^3 + 1.3^3 + 1.5^3 + 1.7^3 + 1.9^3]$

D $0.2 \times [0.1^3 + 0.3^3 + 0.5^3 + 0.7^3 + 0.9^3 + 1.1^3 + 1.3^3 + 1.5^3 + 1.7^3 + 1.9^3]$

E $0.2 \times [0.25^3 + 0.5^3 + 0.75^3 + 1^3 + 1.25^3 + 1.5^3 + 1.75^3 + 2^3]$

[1 mark]

5 The value of the definite integral $\int_0^{\frac{\pi}{6}} \sin(x)\,dx$ is:

A $-\frac{\sqrt{3}}{2}$

B $\frac{2-\sqrt{3}}{2}$

C $\frac{\sqrt{3}-2}{2}$

D $\frac{1}{2}$

E $-\frac{1}{2}$

[1 mark]

Part B

Short-answer questions

Six short-answer questions
56 marks
Show your working where appropriate.

1 Find the approximate area under the curve $f(x) = 2x^3 + 1$ from $x = 1$ to $x = 3$ by using:

a 4 left rectangles

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b 4 right rectangles

c 4 centred rectangles.

[9 marks]

2 a Use right rectangles with width 0.25 units to find an approximation to:

i $\int_0^3 x^2\,dx$

ii $\int_0^2 x^2\,dx$

iii $\int_2^3 x^2\,dx$

b Show that $\int_0^3 x^2\,dx = \int_0^2 x^2\,dx + \int_2^3 x^2\,dx$.

[12 marks]

3 Find an approximation to $\int_1^7 \frac{1}{x}dx$ by using rectangles with width 1 unit:

a left rectangles

b right rectangles

c centred rectangles

[9 marks]

4 Evaluate each of the following definite integrals:

a $\int_1^3 (2x - 9)dx$

b $\int_2^6 e^x \, dx$

c $\int_0^{\pi} \cos(x)dx$

d $\int_{-2}^1 (x^2 - 3x + 5)dx$

[12 marks]

5 a Evaluate $\int_{-3}^3 2x^3 \, dx$.

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b Find the area enclosed between the curve $y = 2x^3$ and the x-axis between $x = -3$ and $x = 3$.

[6 marks]

6 Evaluate:

a $\int_{-1}^{2}(x^2 - 2x + 1)dx + \int_{-1}^{2}(2x^2 + 4x - 3)dx$

b $\int_{0}^{1}(5x^3 - 2x^2 + x - 2)dx - \int_{0}^{1}(x^3 - 5x^2 + 4)dx$

[8 mark]

Part C

Analysis questions

Two analysis questions
19 marks
Show your working where appropriate.

1 a Evaluate $\int_{-1}^{3}(2x + 2)dx$.

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b Shade the area bounded by $y = 2x + 2$ and the x-axis between $x = -1$ and $x = 3$.

c Find the volume of the cone formed if this area is rotated around the x-axis.

d Show that the volume of the cone is given by $V = \pi\int_{-1}^{3}(2x + 2)^2\,dx$.

[12 marks]

2 a Differentiate $\tan(3x)$.

b Find the value of $\int_{\frac{\pi}{12}}^{\frac{\pi}{9}} \frac{1}{\cos^2(3x)}\,dx$.

[7 marks]

[Total marks: 80]

Answers

Part A – Multiple-choice questions [1 mark each]

1 The statement $\int_b^a f(x)dx = \int_a^b f(x)dx$ is not correct since $\int_b^a f(x)dx = -\int_a^b f(x)dx$.

$\therefore$ **C** **[1 mark]**

2 The algebraic area between $x = -4$ and $x = 1$ is negative, so $-\int_{-4}^{1} f(x)dx$ will give the physical area.

$\therefore$ **E** **[1 mark]**

3 Width of each rectangle is 0.5 units.

Heights are $f(0), f(0.5), f(1), f(1.5)$

i.e. $4 - 0^2, 4 - 0.5^2, 4 - 1^2, 4 - 1.5^2$

$$\text{Total area} = 0.5 \times (4 - 0^2) + 0.5 \times (4 - 0.5^2) + 0.5 \times (4 - 1^2) + 0.5 \times (4 - 1.5^2)$$
$$= 0.5 \times [4 - 0^2 + 4 - 0.5^2 + 4 - 1^2 + 4 - 1.5^2$$

$\therefore$ **A** **[1 mark]**

4 The width of each rectangle is 0.2 and the centres are at 0.1, 0.3, 0.5, …, 0.9

The heights are $f(0.1), f(0.3), f(0.5), \ldots, f(1.9)$

$$\text{Total area} = 0.2 \times 0.1^3 + 0.2 \times 0.3^3 + 0.2 \times 0.5^3 + 0.2 \times 0.7^3 + 0.2 \times 0.9^3 + 0.2 \times 1.1^3 + 0.2 \times 1.3^3 + 0.2 \times 1.5^3 + 0.2 \times 1.7^3 + 0.2 \times 1.9^3$$
$$= 0.2 \times [0.1^3 + 0.3^3 + 0.5^3 + 0.7^3 + 0.9^3 + 1.1^3 + 1.3^3 + 1.5^3 + 1.7^3 + 1.9^3]$$

$\therefore$ **D** **[1 mark]**

5 $\int_0^{\frac{\pi}{6}} \sin(x)dx = [-\cos(x)]_0^{\frac{\pi}{6}}$

$$= -\cos\left(\frac{\pi}{6}\right) - [-\cos(0)]$$
$$= -\frac{\sqrt{3}}{2} + 1$$
$$= \frac{2}{2} - \frac{\sqrt{3}}{2}$$
$$= \frac{2 - \sqrt{3}}{2}$$

$\therefore$ **B** **[1 mark]**

Part B – Short-answer questions

1 a Width of each rectangle is 0.5, heights $f(1), f(1.5), f(2), f(2.5)$ **[1 mark]**

$\text{Area} = 0.5 \times [2(1)^3 + 1] + 0.5 \times [2(1.5)^3 + 1] + 0.5 \times [2(2)^3 + 1] + 0.5 \times [2(2.5)^3 + 1]$ **[1 mark]**

$= 30$ **[1 mark]**

b Width of each rectangle is 0.5, heights $f(1.5), f(2), f(2.5), f(3)$ **[1 mark]**

$\text{Area} = 0.5 \times [2(1.5)^3 + 1] + 0.5 \times [2(2)^3 + 1] + 0.5 \times [2(2.5)^3 + 1] + 0.5 \times [2(3)^3 + 1]$ **[1 mark]**

$= 56$ **[1 mark]**

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c Width of each rectangle is 0.5, heights $f(1.25), f(1.75), f(2.25), f(2.75)$ **[1 mark]**

Area $= 0.5 \times [2(1.25)^3 + 1] + 0.5 \times [2(1.75)^3 + 1] + 0.5 \times [2(2.25)^3 + 1]$
$+ 0.5 \times [2(2.75)^3 + 1]$ **[1 mark]**

$= 41.5$ **[1 mark]**

2 a i $A = 0.25 \times [0.25^2 + 0.5^2 + 0.75^2 + 1^2 + 1.25^2 + \ldots + 3^2]$ **[2 marks]**

$= 10.15625$ **[1 mark]**

ii $A = 0.25 \times [0.25^2 + 0.5^2 + 0.75^2 + 1^2 + 1.25^2 + 1.5^2 + 1.75^2 + 2^2]$ **[2 marks]**

$= 3.1875$ **[1 mark]**

iii $A = 0.25 \times [2.25^2 + 2.5^2 + 2.75^2 + 3^2]$ **[2 marks]**

$= 6.96875$ **[1 mark]**

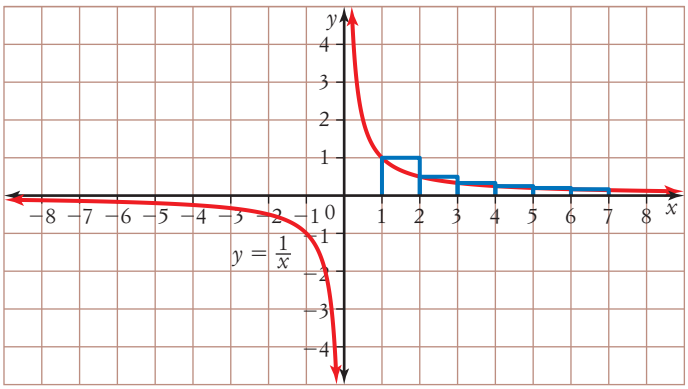
b $\int_0^2 x^2\,dx + \int_2^3 x^2\,dx = 3.1875 + 6.96875$ **[1 mark]**

$= 10.15625$

$= \int_0^3 x^2\,dx$ **[1 mark]**

$\therefore \int_0^3 x^2\,dx = \int_0^2 x^2\,dx + \int_2^3 x^2\,dx$ **[1 mark]**

3 a

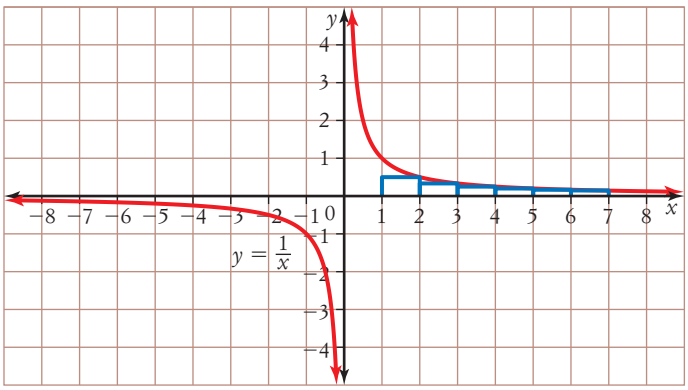


[1 mark]

$A = 1 \times \frac{1}{1} + 1 \times \frac{1}{2} + 1 \times \frac{1}{3} + 1 \times \frac{1}{4} \times 1 \times \frac{1}{5} + 1 \times \frac{1}{6}$ **[1 mark]**

$= 2\frac{9}{20}$ **[1 mark]**

b

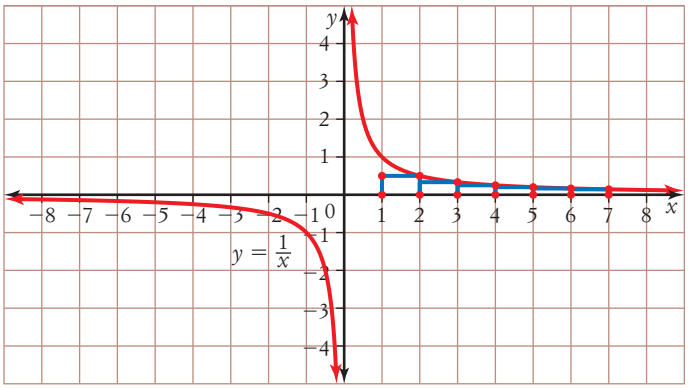


[1 mark]

$A = 1 \times \frac{1}{2} + 1 \times \frac{1}{3} + 1 \times \frac{1}{4} \times 1 \times \frac{1}{5} + 1 \times \frac{1}{6} + 1 \times \frac{1}{7}$ **[1 mark]**

$= 1\ \frac{83}{140}$ **[1 mark]**

c



[1 mark]

$A = 1 \times \frac{1}{1.5} + 1 \times \frac{1}{2.5} + 1 \times \frac{1}{3.5} \times 1 \times \frac{1}{4.5} + 1 \times \frac{1}{5.5} + 1 \times \frac{1}{6.5}$ **[1 mark]**

$= 1.91$ **[1 mark]**

4 a $\int_1^3 (2x - 9)dx = \left[x^2 - 9x\right]_1^3$ **[1 mark]**

$= (3^2 - 9 \times 3) - (1^2 - 9 \times 1)$ **[1 mark]**

$= 9 - 27 - 1 + 9$

$= -10$ **[1 mark]**

b $\int_2^6 e^x\, dx = \left[e^x\right]_2^6$ **[1 mark]**

$= e^6 - e^2$ **[1 mark]**

$= e^2(e^4 - 1)$ **[1 mark]**

c $\int_0^{\pi} \cos(x)dx = [\sin(x)]_0^{\pi}$ **[1 mark]**

$= \sin(\pi) - \sin(0)$ **[1 mark]**

$= 0 - 0$

$= 0$ **[1 mark]**

d $\int_{-2}^{1} (x^2 - 3x + 5)dx = \left[\frac{x^3}{3} - \frac{3x^2}{2} + 5x\right]_{-2}^{1}$ **[1 mark]**

$= \left(\frac{1^3}{3} - \frac{3 \times 1^2}{2} + 5 \times 1\right) - \left(\frac{(-2)^3}{3} - \frac{3(-2)^2}{2} + 5 \times -2\right)$ **[1 mark]**

$= \left(\frac{1}{3} - \frac{3}{2} + 5\right) - \left(\frac{-8}{3} - 6 - 10\right)$

$= \frac{1}{3} - \frac{3}{2} + 5 + \frac{8}{3} + 6 + 10$

$= 22\frac{1}{2}$ **[1 mark]**

5 a $\int_{-3}^{3} 2x^3\,dx = \left[\frac{2x^4}{4}\right]_{-3}^{3}$

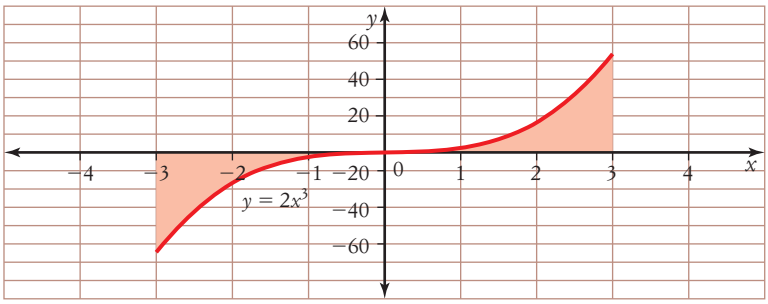
$= \left[\frac{x^4}{2}\right]_{-3}^{3}$ **[1 mark]**

$= \frac{3^4}{2} - \frac{(-3)^4}{2}$ **[1 mark]**

$= \frac{81}{2} - \frac{81}{2}$

$= 0$ **[1 mark]**

b



$A = 2\int_{0}^{3} 2x^3\,dx$ **[1 mark]**

$= \int_{0}^{3} 4x^3\,dx$

$= \left[\frac{4x^4}{4}\right]_{0}^{3}$

$= \left[x^4\right]_{0}^{3}$ **[1 mark]**

$= 3^4 - 0^4$

$= 81$

The area is 81 units^2. **[1 mark]**

6 a $\int_{-1}^{2}(x^2 - 2x + 1)dx + \int_{-1}^{2}(2x^2 + 4x - 3)dx$

$= \int_{-1}^{2}(3x^2 + 2x - 2)dx$ **[1 mark]**

$= \left[x^3 + x^2 - 2x\right]_{-1}^{2}$ **[1 mark]**

$= (2^3 + 2^2 - 2 \times 2) - [(-1)^3 + (-1)^2 - 2 \times (-1)]$ **[1 mark]**

$= 8 + 4 - 4 + 1 - 1 - 2$

$= 6$ **[1 mark]**

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b $\int_0^1 (5x^3 - 2x^2 + x - 2)dx - \int_0^1 (x^3 - 5x^2 + 4)dx$

$= \int_0^1 (4x^3 + 3x^2 + x - 6)dx$ **[1 mark]**

$= \left[x^4 + x^3 + \frac{x^2}{2} - 6x\right]_0^1$ **[1 mark]**

$= (1^4 + 1^3 + \frac{1^2}{2} - 6 \times 1) - (0^4 + 0^3 + \frac{0^2}{2} - 6 \times 0)$ **[1 mark]**

$= 1 + 1 + \frac{1}{2} - 6 - 0$

$= -3\frac{1}{2}$ **[1 mark]**

Part C – Analysis questions

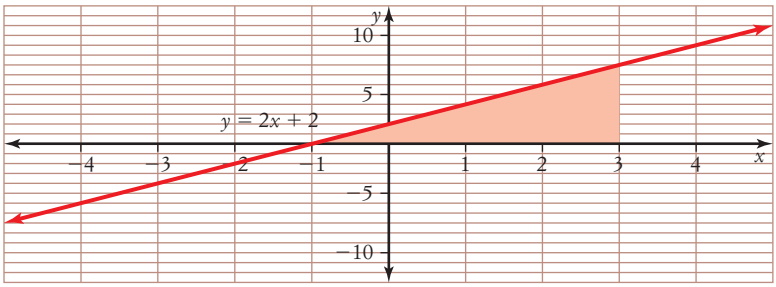
1 a $\int_{-1}^3 (2x + 2)dx = \left[x^2 + 2x\right]_{-1}^3$ **[1 mark]**

$= (3^2 + 2 \times 3) - [(-1)^2 + 2 \times (-1)]$ **[1 mark]**

$= 9 + 6 - 1 + 2$

$= 16$ **[1 mark]**

b



[1 mark]

c $\pi \int_{-1}^3 (2x + 2)^2 dx$

$= \pi \int_{-1}^3 2^2 (x + 1)^2 dx$

$= 4\pi \int_{-1}^3 (x + 1)^2 dx$

$= \frac{4\pi}{3}\left[(x + 1)^3\right]_{-1}^3$

$= \frac{4\pi}{3}(4^3 - 0)$

$= \frac{256\pi}{3}$ **[1 mark]**

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d $\pi\int_{-1}^{3}(2x+2)^2\,dx = \pi\int_{-1}^{3}(4x^2+8x+4)^2\,dx$

$= \pi\left[\frac{4x^3}{3}+\frac{8x^2}{2}+4x\right]_{-1}^{3}$

$= \pi\left[\frac{4x^3}{3}+4x^2+4x\right]_{-1}^{3}$ **[1 mark]**

$= \pi\left[\left(\frac{4\times 3^3}{3}+4\times 3^2+4\times 3\right)-\left(\frac{4\times(-1)^3}{3}+4\times(-1)^2+4\times(-1)\right)\right]$ **[1 mark]**

$= \pi\left[(36+36+12)-\left(-\frac{4}{3}+4-4\right)\right]$

$= \frac{256\pi}{3}$ **[1 mark]**

From part **c**, this is the volume of the cone.

So $V = \pi\int_{-1}^{3}(2x+2)^2\,dx$ gives the volume of the cone. **[1 mark]**

2 a $\frac{dy}{dx} = 3\times\frac{1}{\cos^2(3x)}$ **[1 mark]**

$= \frac{3}{\cos^2(3x)}$ **[1 mark]**

b $\int_{\frac{\pi}{12}}^{\frac{\pi}{9}}\frac{1}{\cos^2(3x)}dx = \int_{\frac{\pi}{12}}^{\frac{\pi}{9}}\frac{1}{3}\times\frac{3}{\cos^2(3x)}dx$ **[1 mark]**

$= \frac{1}{3}\int_{\frac{\pi}{12}}^{\frac{\pi}{9}}\frac{3}{\cos^2(3x)}dx$ **[1 mark]**

$= \frac{1}{3}[\tan(3x)]_{\frac{\pi}{12}}^{\frac{\pi}{9}}$ **[1 mark]**

$= \frac{1}{3}\left[\tan\left(3\times\frac{\pi}{9}\right)-\tan\left(3\times\frac{\pi}{12}\right)\right]$ **[1 mark]**

$= \frac{1}{3}\left[\tan\left(\frac{\pi}{3}\right)-\tan\left(\frac{\pi}{4}\right)\right]$

$= \frac{1}{3}\left[\sqrt{3}-1\right]$ **[1 mark]**

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